

ASMEETA PRAKASH SAYAJI

Quantum Computing Theory — Noise, Verification, and Complexity

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github.com/asmeeta-quantum/ETH_Qiskit_Workspace

RESEARCH PROFILE

Research interest in the theoretical foundations of quantum computing, focusing on how realistic noise processes influence computational power, verifiability, and classical simulability boundaries. Combines analytical modeling of open quantum systems with numerical simulations using Qiskit and QuTiP. Background at ISRO provides practical insight into physical noise sources in cryogenic and microwave systems relevant for quantum devices. Research goal: develop rigorous frameworks for verification complexity, noise-adapted quantum advantage thresholds, and resource-theoretic descriptions of noisy quantum computation.

KEY RESEARCH CONTRIBUTIONS

- **Verification Complexity under Noise:** Derived analytical scaling bounds for verification sample complexity under T_1/T_2 relaxation and dephasing channels, identifying regimes where quantum advantage becomes classically simulable
- **Error Mitigation Theory and Simulation:** Implemented Zero-Noise Extrapolation (ZNE) framework demonstrating $\approx 20\%$ fidelity improvement in noisy quantum circuits through systematic noise amplification analysis
- **Quantum Control Optimization:** Designed DRAG pulse optimization framework achieving $> 90\%$ suppression of $|1\rangle \rightarrow |2\rangle$ leakage in transmon qubits, linking control theory to noise suppression limits

RESEARCH PORTFOLIO

Complete Workspace Structure

```
ETH_Qiskit_Workspace/  
01_bell_states_entanglement.ipynb  
02_statevector_simulation.ipynb  
03_depolarizing_noise_analysis.ipynb  
04_superconducting_noise_channels.ipynb (T1/T2 modeling for transmons)  
05_composite_noise_simulation.ipynb  
06_grover_verification.ipynb  
07_deutsch_jozsa_advantage.ipynb  
08_verification_sample_complexity.ipynb (Key result: analytical bounds)  
09_lindblad_qubit_dynamics.ipynb  
10_drag_pulse_optimization.ipynb (Key result: > 90% suppression)  
11_noise_bench_qc/ # Benchmarking suite  
12_lindblad_sim/ # Simulation toolkit
```

Highlighted Theoretical Components

- Lindblad master equation simulations for open quantum system dynamics
- Noise channel modeling (amplitude damping, phase damping, depolarizing noise)
- Analytical verification bounds for noisy quantum circuits
- Simulation of algorithmic advantage under realistic noise assumptions

Full repository: https://github.com/asmeeta-quantum/ETH_Qiskit_Workspace

TECHNICAL SKILLS

Quantum Theory	Open quantum systems, Lindblad dynamics, quantum channels, verification protocols, entropy measures, resource theories, noise modeling
Quantum Software	Qiskit (Terra, Aer, Pulse), QuTiP, OpenQASM
Programming	Python (NumPy, SciPy, Matplotlib), Git, LaTeX
Hardware	Cryogenic systems (4K), microwave instrumentation, vector network analyzer measurements, phase noise characterization, signal stability analysis

PROFESSIONAL EXPERIENCE

Space Applications Centre (ISRO) — Senior Assistant

2014–Present

- Evaluate technical specifications for cryogenic and microwave components used in space science payloads operating near 4K
- Analyze phase noise, amplifier noise figures, and S-parameter measurements from VNA characterization reports
- Interpret hardware noise characteristics and connect experimental parameters to abstract quantum noise channel models
- Develop technical understanding of stability constraints relevant to quantum device environments

EDUCATION

M.Sc. Computer Science

North Maharashtra University (2009)

- Relevant preparation: linear algebra, probability theory, algorithms, information theory
- Independent study: quantum information theory (Nielsen & Chuang), open quantum systems (Breuer & Petruccione), functional analysis foundations

B.Sc. Computer Science, Physics & Mathematics

North Maharashtra University (2006)

- Rank: 1/80
- Physics preparation: quantum mechanics, classical mechanics, mathematical methods
- Mathematics preparation: linear algebra, differential equations, numerical methods
- Computer science preparation: programming, discrete mathematics

RESEARCH INTERESTS

- Complexity limits of noisy quantum computation
- Verification and certification of quantum computations
- Classical simulability thresholds under realistic noise
- Lindbladian models of open quantum systems
- Error mitigation limits and theoretical guarantees
- Resource theories for noisy quantum information processing

ACADEMIC ACHIEVEMENTS

- **Rank 1** in B.Sc. Computer Science
- **JNU Merit Scholarship** for academic excellence
- Research Outcome: $> 90\%$ leakage suppression in quantum control experiments
- Research Outcome: $\approx 20\%$ fidelity improvement via zero-noise extrapolation
- Research Outcome: Analytical bounds for verification complexity under T_1/T_2 noise

RESEARCH OBJECTIVE

PhD research in theoretical quantum computing focusing on mathematically rigorous understanding of noise effects on computational advantage, verification complexity, and scalability of quantum algorithms in realistic physical regimes.